



North South University
Department of Electrical & Computer Engineering

Fall 2025

Course Title: Analog Electronics I Lab

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Section: 7

Experiment Number: 02

Experiment Name:

Diode Rectifier Circuits

Experiment Date: 21.10.2025

Date of Submission: 28.10.2025

Group Number: 01

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Experiment name: Diode rectifier circuits.

Objective: This experiment aims,

- To study different diode rectifier circuits.
- To study the working principle and operation of half-wave, full-wave and bridge rectifier circuits.
- To understand how diode rectifier circuits convert AC voltage into DC voltage.

Theory: A rectifier converts an AC signal into a DC signal using a diode, which conducts current only in forward bias and remains open in reverse bias. When an alternating voltage is applied, the diode allows only the positive half cycle to pass (depending on its orientation), while the negative half cycle will be clipped off. In the output the load will get DC signal.

Diode rectifier can be characterized in two major types. They are :

① Half-wave rectifier

② Full-wave rectifier

Half-wave rectifier: A half-wave rectifier is a circuit that converts AC to DC using a single diode. It allows only one half cycle of the AC signal (Positive or negative) to pass through while blocking the other half, producing a pulsating DC output.

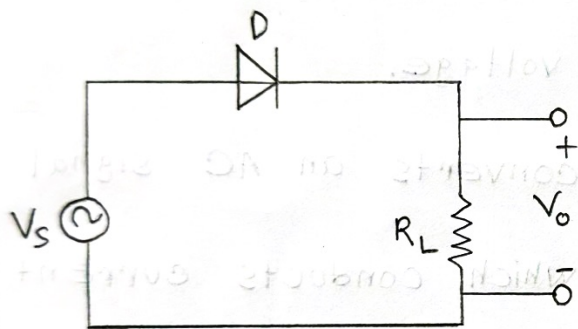


Fig 1: Half-wave rectifier circuit diagram

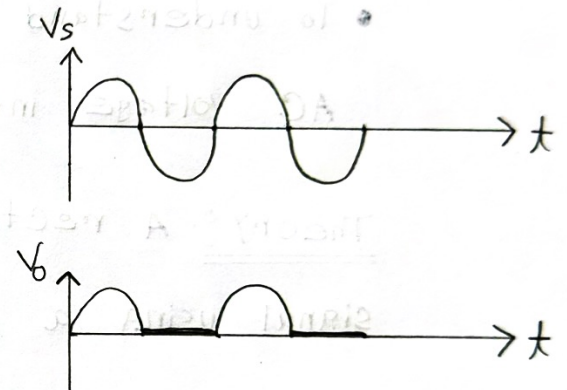


Fig 2: Half-wave rectifier wave shape

The major disadvantages of half-wave rectifier are :

- ① The load receives only about half of the input power.
- ② The average DC output voltage is low.
- ③ The output voltage is not smooth because of the high ripple content.

Full-wave rectifier: A full-wave rectifier is a circuit that converts the entire AC signal into DC by using both halves of the input waveform. It allows current to flow during both positive and negative cycles, producing a smoother and higher average DC output compared to a half-wave rectifier.

Two circuits are used as full-wave rectifier. They are: ① Full-wave rectifier using center-tapped transformer and ② Full-wave bridge rectifier.

Full wave rectifier using center-tapped transformer:

Here, two diodes will be connected to the ends of the transformer and the load will be between the diode and center tap.

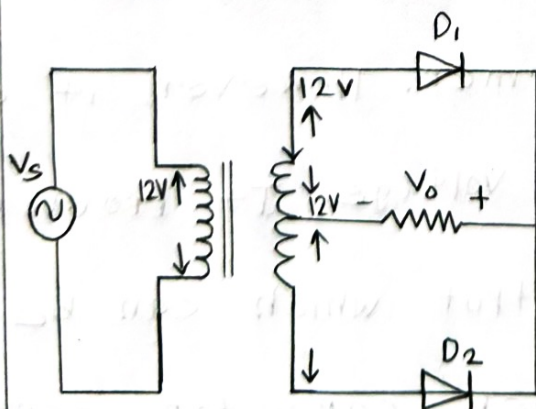


Fig 3: Full-wave rectifier using center-tapped transformer circuit diagram

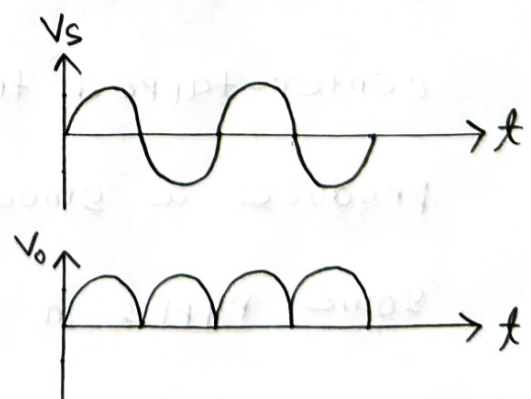


Fig 4: Full-wave rectifier using center-tapped transformer wave shape.

The advantages of full-wave rectifier using center-tapped transformer are:

- ① Power loss is reduced.
- ② The average DC output voltage is considerably higher.
- ③ The output waveform is smoother.

The drawbacks of full-wave rectifier using center-tapped transformer are:

- ① It occupies more space and becomes bulky due to the transformer.
- ② It is not cost effective because of the transformer.

Full-wave bridge rectifier: Here, four diodes will be connected as bridge connection. It overcomes all the disadvantages of full-wave rectifier using center-tapped transformer. However, it cannot produce a smooth DC voltage. It produces some ripple in the output which can be reduced by using filter capacitor across the load.

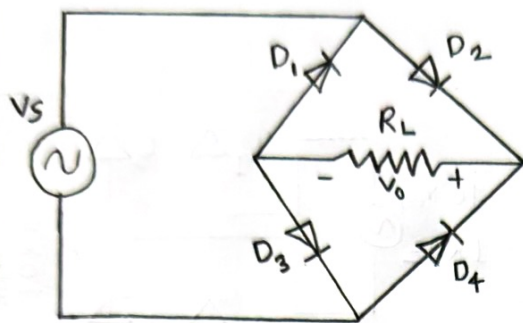


Fig 5: Full-Wave Bridge rectifier circuit diagram

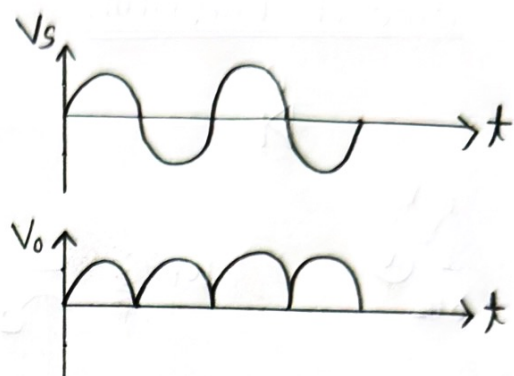


Fig 6: Full-Wave Bridge rectifier wave shape

Equipment List:

Serial no.	Component Details	Specification	Quantity
1.	P-n junction diode	IN4007	4 Piece
2.	Resistor	10k Ω	1 piece
3.	Capacitor	0.22 MF, 10 μ F	1 Piece each
4.	Signal Generator		1 unit
5.	Trainer Board		1 unit
6.	Oscilloscope		1 unit
7.	Digital Multimeter		1 Unit
8.	Chords and Wire		As required

Circuit Diagram:

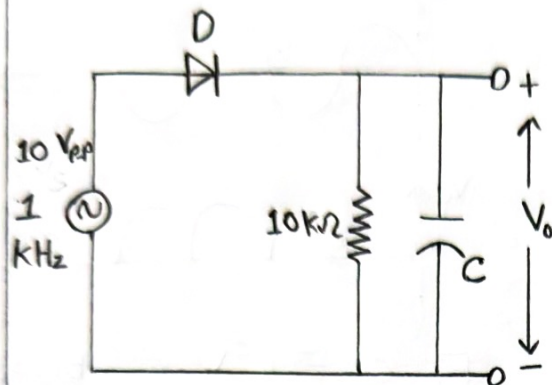


Fig 7: Half-Wave
Circuit Diagram

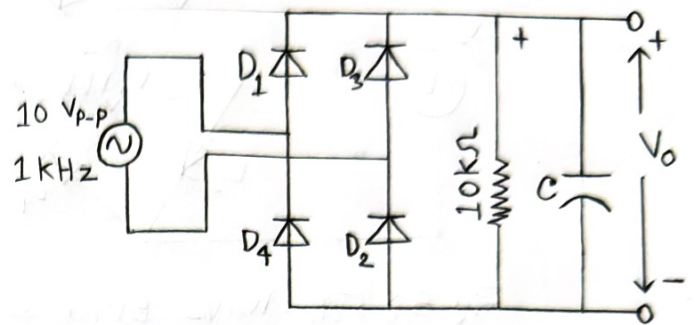


Fig 8: Full-Wave Circuit
Diagram

Data Table:

Theoretical Value of $R = 10k\Omega$

Measured value of $R = 9.81k\Omega$

	Half Wave Rectifier	Full Wave Rectifier
V_o (P-P) (Without capacitor)	4.80V	4.32V
V_o (P-P) (With 0.22 μF)	1.60V	640 mV
V_o (P-P) (With 10 μF)	320 mV	160 mV

Question and Answer :

1. For Fig 2.4 and Fig 2.5, draw the input-output wave-shape without capacitor, with $0.22 \mu F$ capacitor and $10 \mu F$ capacitor.

Ans: Graphs are attached.

2. Compare the change in the wave-shape and peak to peak values for no capacitor at the output to $0.22 \mu F$ to $10 \mu F$.

Ans: Comparison of the change in the wave-shape and peak to peak voltage for no capacitor at the output to $0.22 \mu F$ to $10 \mu F$ is given below: (Half-wave rectifier)

Observation	No Capacitor	$0.22 \mu F$	$10 \mu F$
Wave-Shape	Output is a Pulsating DC containing only the positive half cycles. The waveform drops to zero after every cycle producing large ripples.	The capacitor charges at peak and discharges slowly. So, the waveform becomes smoother with small ripples between peaks.	The capacitor maintains constant voltage between peaks, giving a steady DC output.

Observation	No capacitor	0.22 μ F	10 μ F
Peak-to-peak Voltage	4.80V - Very high ripple because voltage falls from maximum to zero in each cycle.	1.60V - Ripple decreases as the capacitor smooths out voltage drops.	320 mV - Capacitor keeps voltage almost constant between cycles.

Comparison of the change in the wave-shape and peak-to-peak voltage for no capacitor at the output to 0.22 μ F to 10 μ F is given below :

(Full-wave rectifier)

Observation	No Capacitor	0.22 μ F	10 μ F
Wave-shape	Output is pulsating DC with both positive half cycles. Ripple is present but smaller than in the half-wave rectifier.	The capacitor smooths out the waveform. Ripples are reduced and the output becomes more uniform.	The waveform becomes almost pure DC as voltage remains nearly steady throughout the cycle.
peak-to-peak voltage	4.32 V - noticeable ripple because voltage still fluctuates between peaks.	640 mV - Ripple decreases as capacitor filters more effectively between each cycle.	160 mV - The capacitor holds charge efficiently maintaining constant DC output.

3. Explain the effect on the output signal for changing the frequency of the input signal.

Ans:

When the input frequency increases, the output becomes smoother because the capacitor gets charged more often and does not have much time to discharge. This means ripple voltage decreases and the output looks more like steady DC.

When the input frequency decreases, the time between peaks become longer. So, the capacitor discharges more before it recharges again. This causes more ripple and makes the output less smooth.

So, Higher frequency $\rightarrow$ Smoother DC output, less ripple.

Lower frequency $\rightarrow$ Less steady output, more ripple.

4. Between half wave and full wave which circuit produces smoother output? Briefly explain in context with your data collection.

Ans:

Between half-wave and full-wave circuit, the full-wave circuit produces a smoother output.

From the data table, we can see that, the half-wave rectifier, the peak-to-peak voltage decreases from 4.80 V to 1.60 V to 320 mV as the capacitor increases.

And, in the full-wave rectifier, the peak-to-peak voltage decreases from 4.32 V to 640 mV to 160 mV for the same capacitor values.

This shows that the full-wave rectifier has smaller ripple voltages at every capacitance level. It provides a smoother DC output because it rectifies both halves of the AC cycle, allowing the capacitor to recharge more often, reducing the voltage drop between peaks.

So, full-wave circuit produces smoother and more stable DC output than the half-wave circuit.

Discussion:

From this experiment, we learned how half-wave and full-wave rectifier circuits convert alternating current (AC) into direct current (DC) and how adding a filter capacitor affects the output signal. We also observed that increasing the capacitor value significantly reduces the ripple voltage, and the full-wave rectifier provides a smoother DC output than the half-wave rectifier.

From the collected data and oscilloscope, we saw that as the capacitance increased from $0.1\mu\text{F}$ to $10\mu\text{F}$, the output ripple voltage decreased significantly in both rectifier circuits.

We saw, in the half-wave rectifier, only the positive half cycles appear, creating a pulsating DC with large ripples. When a capacitor is added, the waveform becomes smoother as it stores and releases charge between cycles.

And in the full-wave rectifier, both halves of the AC signal are converted to positive cycles, so the output frequency doubles and the waveform is smoother even without a large capacitor. With higher capacitance, the ripple almost disappears, and the output becomes steady DC.

So, we can say that, the collected data and waveforms were mostly similar to what theory predicted; the full-wave rectifier produced smoother output with smaller ripples compared to half-wave rectifier.

During the experiment, we faced difficulties while using the oscilloscope. However, the readings and waveforms were stable throughout the experiment.

In conclusion, the experiment successfully demonstrated the the theoretical concepts of half-wave and full-wave rectifiers.

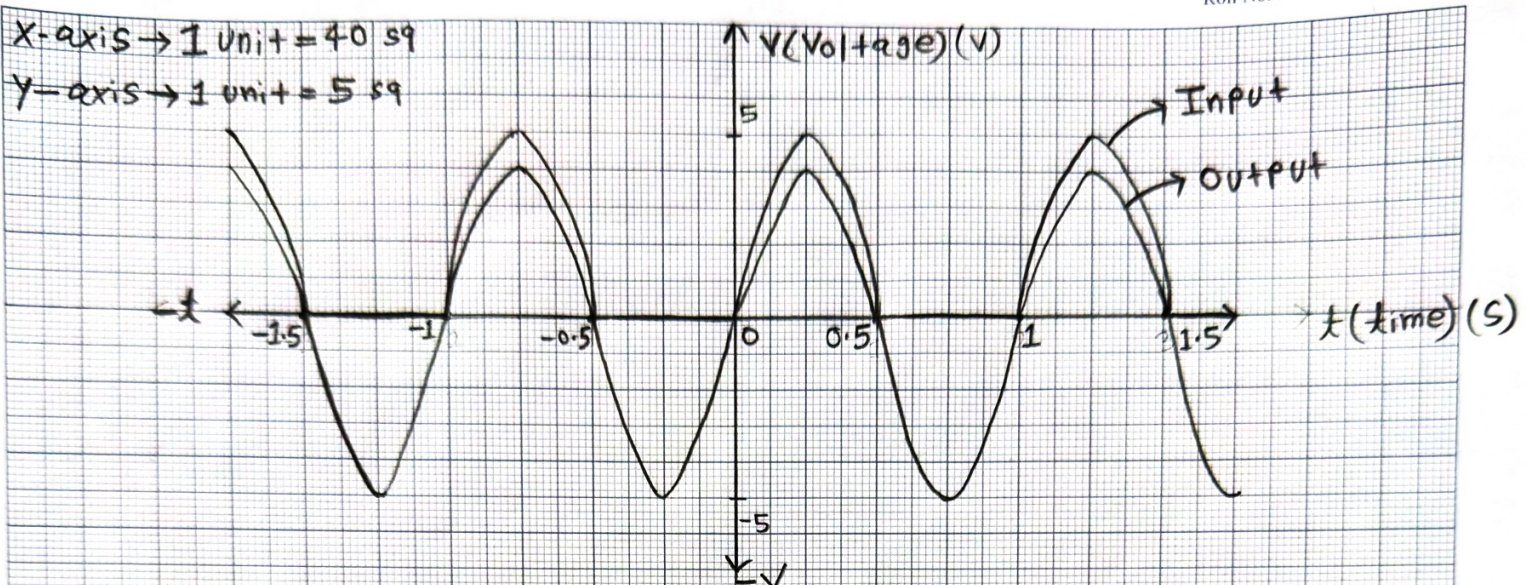


Fig 2.4(a): Input-Output without Capacitor

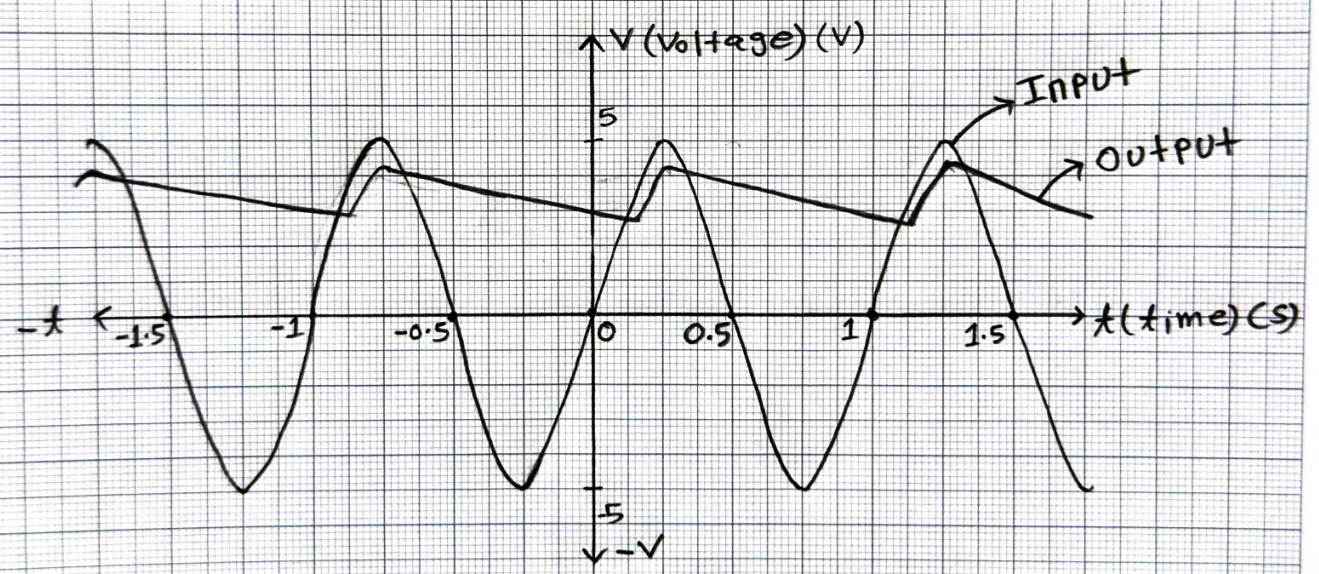


Fig 2.4(b): Input-Output with $0.22 \mu\text{F}$ capacitor

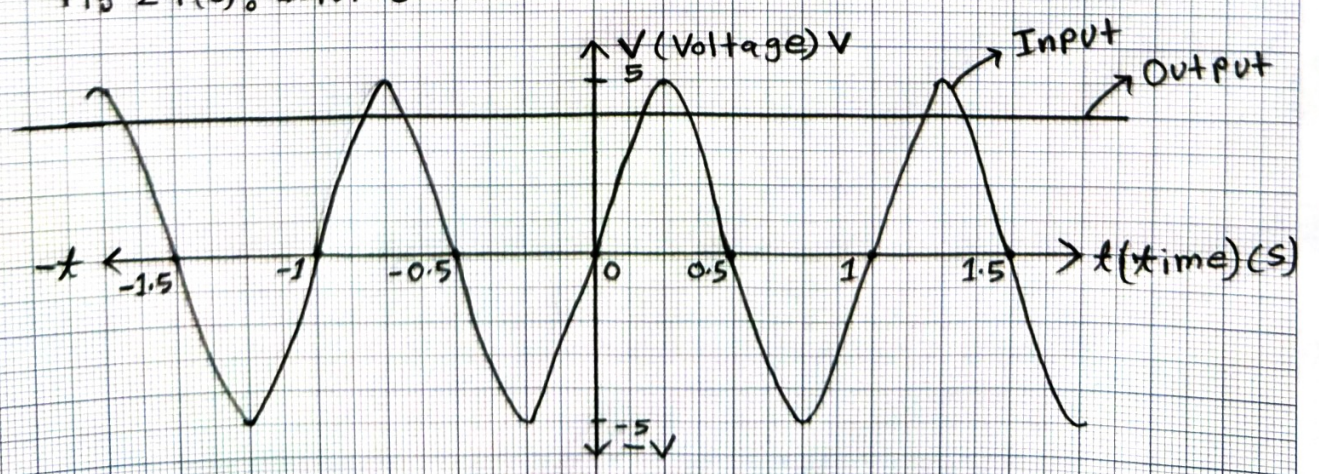


Fig 2.4(c): Input-Output with $10 \mu\text{F}$ capacitor

Fig: Half-Wave Rectifier Input-Output Wave-Shapes

X-axis $\rightarrow 1 \text{ unit} = 40 \text{ s}$
 Y-axis $\rightarrow 1 \text{ unit} = 5 \text{ V}$

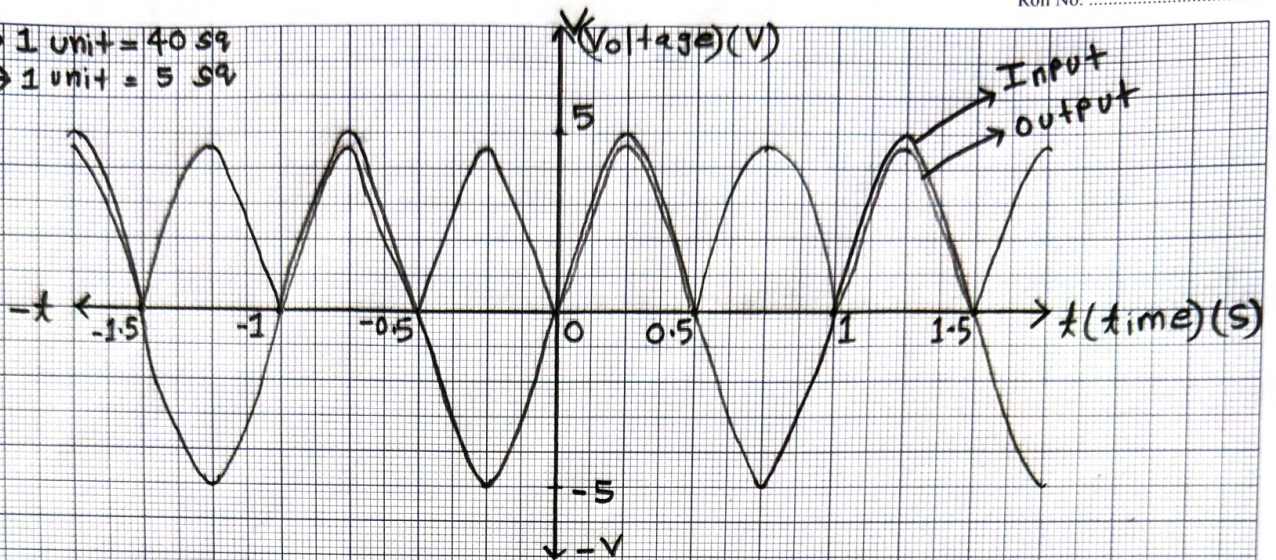


Fig 2.5(d): Input-Output without capacitor

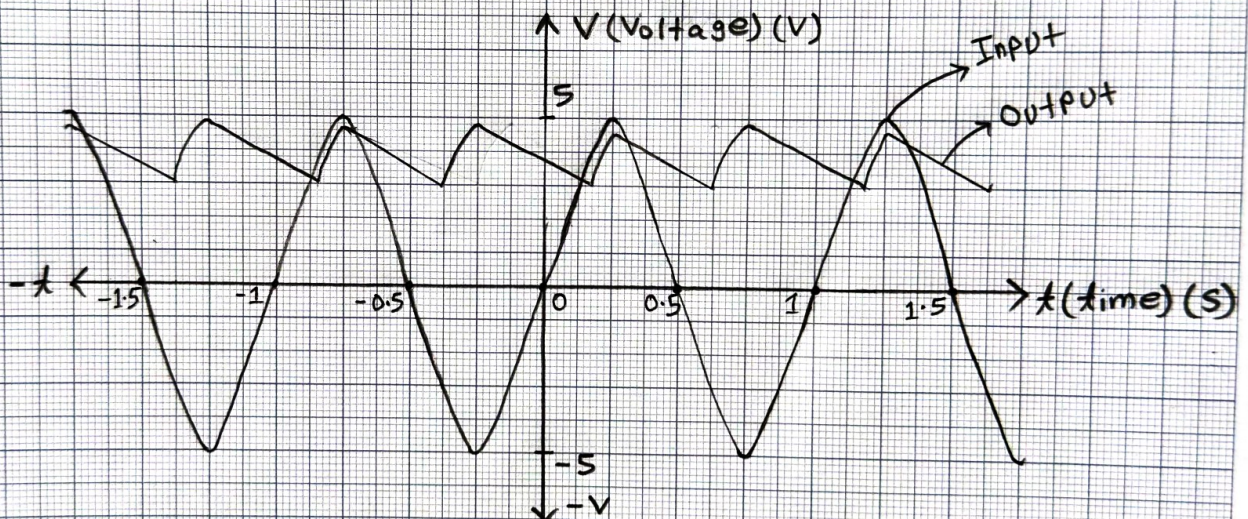


Fig 2.5(e): Input-Output with $0.22\mu\text{F}$ capacitor

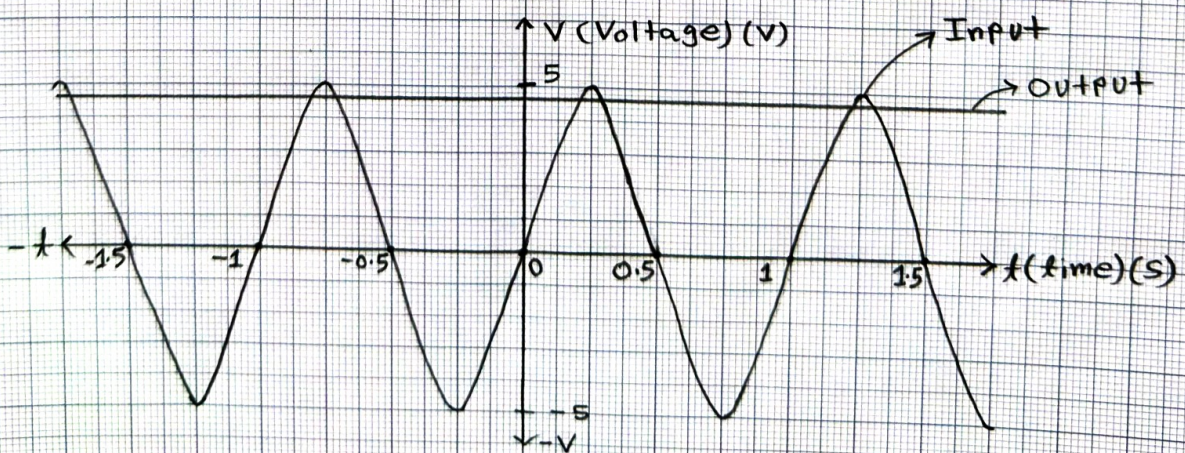


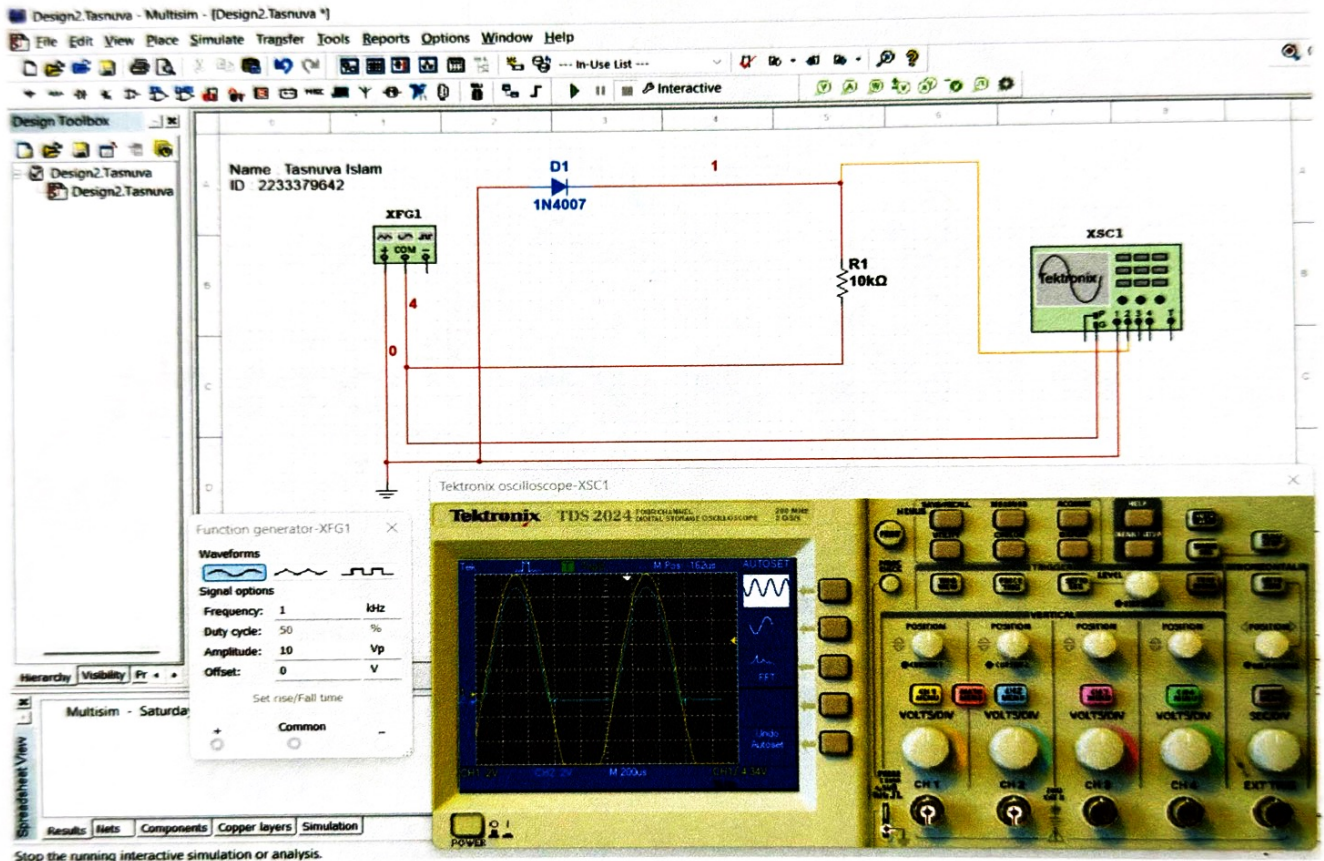
Fig 2.5(f): Input-Output with $10\mu\text{F}$ capacitor

Fig: Full-Wave Rectifier Input-Output Wave-shapes

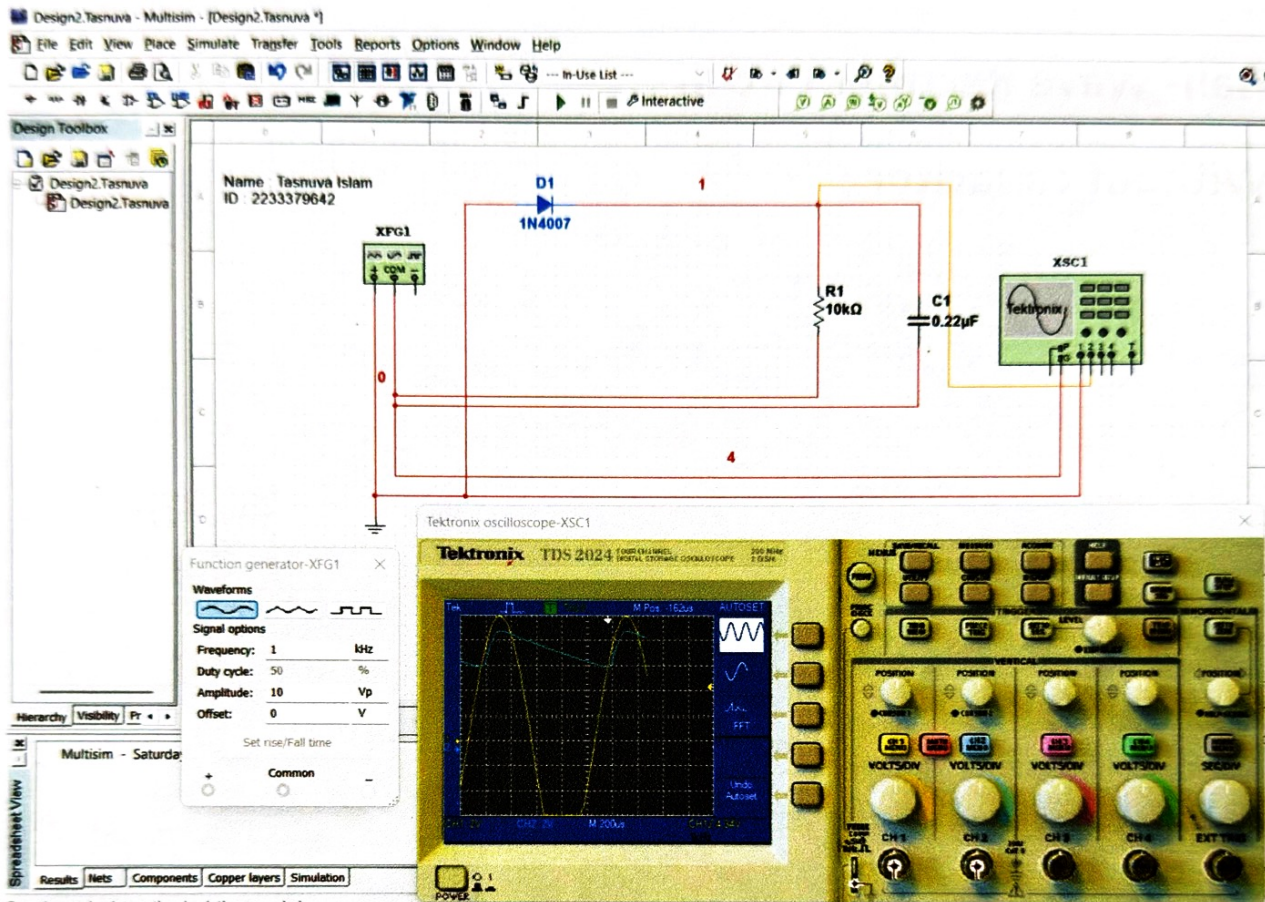
Simulation

Half- Wave Rectifier Circuit :

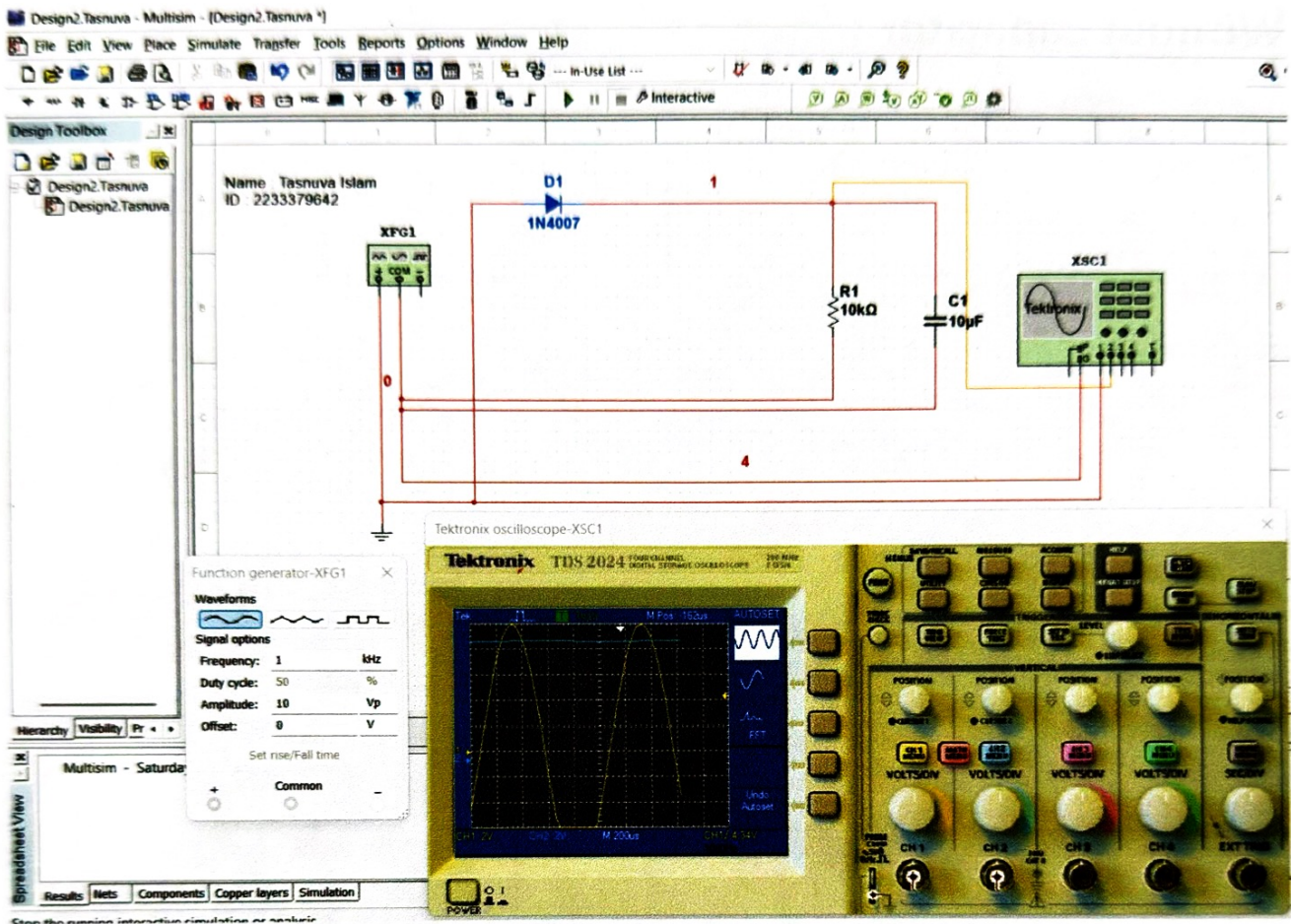
Without capacitor :



With 0.22uF capacitor :

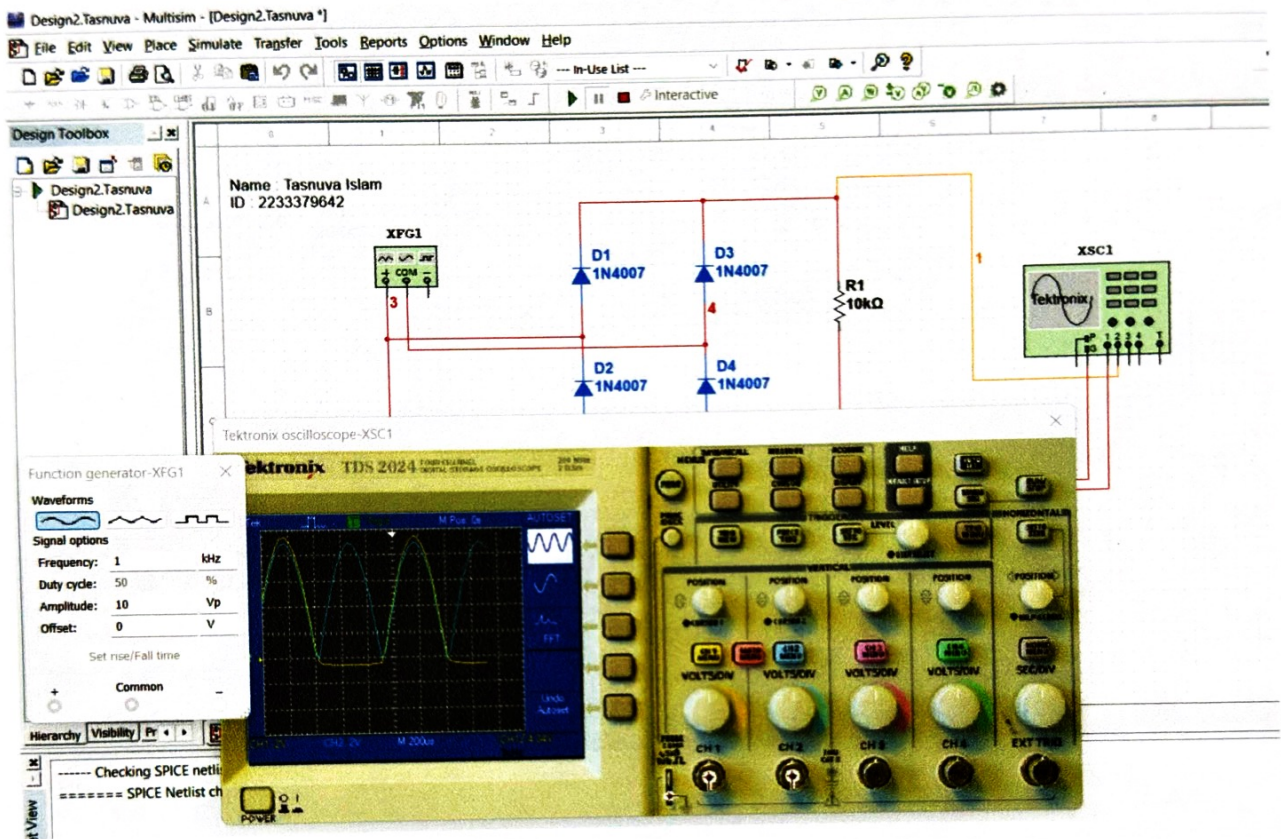


With 10uF capacitor :



Full - Wave Rectifier Circuit :

Without capacitor :



With 0.22uF capacitor :

